Global Ripple Effects of Corporate Tax Reforms

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Motivation

Multinational Enterprises (MNEs) shift large portions of their profits to tax havens:

- Lost corporate income tax (CIT) revenue (% of global CIT): 10% by Tørsløv, Wier and Zucman (2023), 4%–10% by OECD (2015)

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- Pillar 2 GMT (Jan 2024): the multilateral response, in force in the EU and beyond

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Research questions:

1. What are the global effects of these regional tax reforms?
2. How does the multilateral GMT interact with the unilateral TCJA under Side-by-Side?

What we do

1. Develop a multi-country GE model to assess the global effects of the TCJA and GMT:
   - $\rightarrow$ main features of the model: non-rivalry and mobile ownership of intangible capital
   - $\rightarrow$ link intangible investment decisions to profit shifting and tax policies
   - $\rightarrow$ map the institutional detail into the model: territorial shift, bonus depreciation, GILTI, FDII; the GMT's QDMTT/IIR/UTPR hierarchy; the QBAI and SBIE carve-outs

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   - $\rightarrow$ TCJA in the U.S., built up provision by provision
   - $\rightarrow$ GMT everywhere but the U.S., under two regimes for U.S. MNEs: complete exemption vs. local QDMTT top-ups ($=$ Side-by-Side)

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3. Empirical validation of the effects of TCJA on firm investment and profit shifting
   - $\rightarrow$ event studies exploiting between-firm variation in a simulated, pre-determined tax change

What we find

1. TCJA: a large domestic expansion at a fiscal cost, validated by firm-level evidence:
   - $\rightarrow$ U.S. GDP $\uparrow$ 2.1%, driven by investment in tangible and intangible capital; CIT revenue $\downarrow$ 35%
   - $\rightarrow$ positive outward ripple effects on Europe and the RoW

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2. GMT outside the U.S.: CIT revenue $\uparrow$, real activity $\downarrow$ where it binds, most of all in the low-tax region (GDP $\downarrow$ 6.2%):
   - $\rightarrow$ inward ripple effects on the U.S. hinge on the Side-by-Side detail: ≈ 0 if U.S. MNEs are fully exempt; slightly negative once local QDMTTs apply

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3. Scaled by the induced changes in effective tax rates, the two reforms' ripple effects are comparable:
   - $\rightarrow$ $\Rightarrow$ there is no such thing as a purely domestic corporate tax policy

The Model

Model: environment and technology households final good market clearing

- Multi-country GE model:
  - $\rightarrow$ Five regions: US, Europe, Rest of the World, Low Tax (LT), (unproductive) Tax Haven (TH)
  - $\rightarrow$ Representative households; monopolistic competition; governments consume tax revenue (G_i, held fixed)

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- Heterogeneous firms pay fixed costs to export and open subsidiaries (Helpman, Melitz and Yeaple 2004); a firm ω from region i produces in region j:

$$y_{ij}(\omega) = \sigma_{ij} A_j a(\omega)\, z_i(\omega)^{\phi}\, k_{ij}(\omega)^{\alpha}\, \ell_{ij}(\omega)^{1-\phi-\alpha}$$

  - → z_i: nonrival intangible capital (McGrattan and Prescott 2009, 2010), produced at the HQ: $z_i(\omega) = A_i l_i^z(\omega)$
  - → k_{ij}: tangible capital, rented locally; ℓ_{ij}: labor; $\sigma_{ij} \in [0, 1]$: FDI barriers, $\sigma_{ii} = 1$

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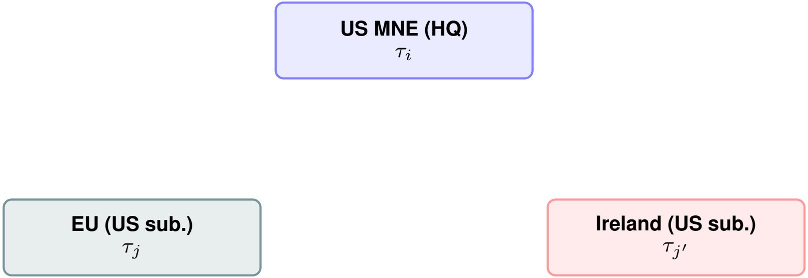
- Profit shifting: transfer the ownership of z to low-tax affiliates at a (mis)priced royalty (Dyrda, Hong and Steinberg 2024) `details`
- Sensitivity: FDI technology spillovers in intangible production `spillovers` `sensitivity`

Profit shifting via transfer pricing: an example



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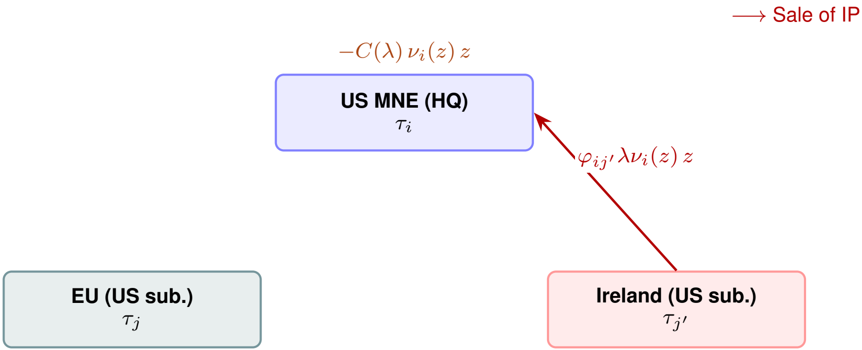

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- $\rightarrow$ $\nu_i(z)\,z$: total licensing fees across the conglomerate
- $\rightarrow$ $\varphi_{ij'} < 1$: markdown of the transfer price
- $\rightarrow$ $C(\lambda)$: profit shifting cost per unit value of the intangible capital paid by the HQ

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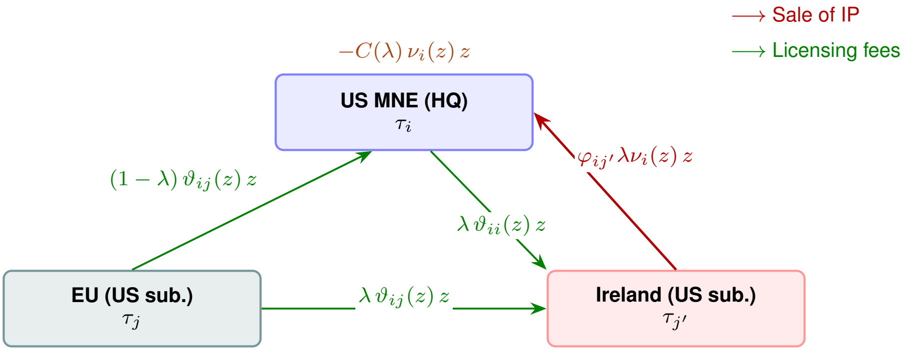

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- $\rightarrow$ $\vartheta_{ij}(z)\,z$: licensing fee paid by the affiliate in j, set $\vartheta_{ij}(z)$ to be the marginal revenue product of z

Firm's problem

- The firm maximizes worldwide after-tax profits (omit firm identifier ω):

$$\max \left\{ \underbrace{\pi_{ii} - T_{ii}}_{\text{After-tax profits in HQ}} + \underbrace{\sum_{j \in J_F} (\pi_{ij} - T_{ij})}_{\text{After-tax profits in productive subsidiaries}} + \underbrace{\mathbb{1}_{\{\lambda_{TH} > 0\}} (\pi_{i,TH} - T_{i,TH})}_{\text{After-tax profits in TH}} \right\}$$

where

→ π_{ij}: the pre-tax profits of a region-i firm's subsidiary in region j details

→ T_{ij}: the tax liability of a region-i firm's subsidiary in region j (policy lever)

→ Taxes are levied on taxable income $\tilde{\pi}_{ij} = \pi_{ij} + (1 - b_j)\, r_j k_{ij}$, with bonus depreciation rate b_j details

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- Two stages of the firm's problem:
  1. Scale choice: given export and FDI destinations and intangible inv., choose rival factors details
  2. Scope choice: choose export and FDI destinations, intangible inv, and profit shifting details

Profit shifting increases intangible investment

Assume $\mathcal{C}_{i,j}(\lambda) = \psi_{i,j}\left[\lambda + (1-\lambda)\log(1-\lambda)\right]$. Then, in a territorial tax system:

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$\rightarrow$ λ_{LT} $\nearrow$ in tax differential $(\tau_i - \tau_{LT})$ and $\nearrow$ in mispricing the intangible capital $(1-\varphi_{i,LT})$, $\searrow$ in the shifting cost $\psi_{i,LT}$

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- Intangible investment:

$$z = z^{NS} \times \underbrace{\left(1 + W_i\left(\lambda\mathcal{C}'(\lambda) - \mathcal{C}(\lambda)\right)\right)^{1+\phi(\varrho-1)}}_{\Lambda(\lambda)\geq 1}$$

  → z^{NS}: optimal intangible investment when firms do not shift profits, i.e. $\lambda = 0$

  → $\Lambda(\lambda) \geq 1$: net gain from profit shifting per unit of intangible capital, $\nearrow$ in λ (Albertus et al. 2019)

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Key takeaway: profit shifting increases the after-tax return of intangible investment.

Tax Reforms: TCJA and GMT

Summary of the corporate tax reforms

Pre-TCJA GILTI/FDII Bonus OBBBA Other GMT STTR/SBIE

		TCJA				
	Pre-TCJA	**Rate+terr.**	**Bonus depr.**	**GILTI**	**FDII**	**GMT**
Regime	Worldwide	Quasi-territorial	Immediate expensing of tangible investment	*Stick*: **tax** foreign intan. income	*Carrot*: **subsidize** domestic intan. income	Top-up tax on undertaxed income
Tax rate	35%	21%	Expensing: 50% $\rightarrow$ 100%	10.5%	13.125%	15%
Tax base	Worldwide income; foreign income until repatriation	Domestic income + GILTI	Qualifying U.S. tangible inv.; eligible share $0.80 \rightarrow 0.66$	Net foreign income with high-tax exclusion	Export net of allocable costs	Foreign income with ETR $<$ 15%
Exclusion	–	–	–	**QBAI**: 10% return on for. tang.	**QBAI**: 10% return on dom. tang.	**SBIE**: 5% tangibles $+$ 5% payroll
Foreign tax credit	Full FTC on repatriated earnings	–	–	80% FTC on tax paid on the base	–	Full credit on paid taxes

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OBBBA (2025): 100% expensing made permanent; GILTI $\to$ NCTI (12.6%), FDII $\to$ FDDEI (14%), QBAI exclusions repealed.

Taking the Model to the Data

Calibration `Measure PS` `Parameters`

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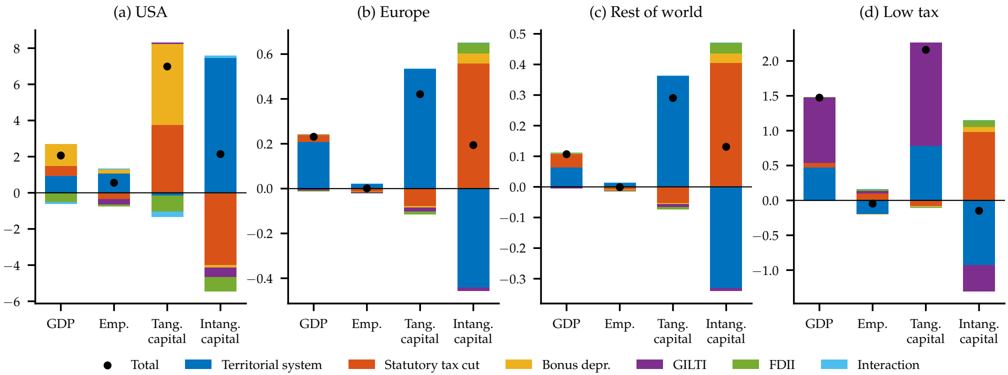
- Profit-shifting costs: lost profits from TWZ (2023): 0.8% of GDP (US), 1.2% (Europe), 1.4% (RoW)

- Repatriation share from LT/TH (ι_L): calibrated inside the TCJA experiment to match the -48% change in aggregate lost profits (Delis, Delis, Laeven and Ongena 2024)

Counterfactual experiments: TCJA

Global effects of TCJA: GDP and investment

details by firm type incentives



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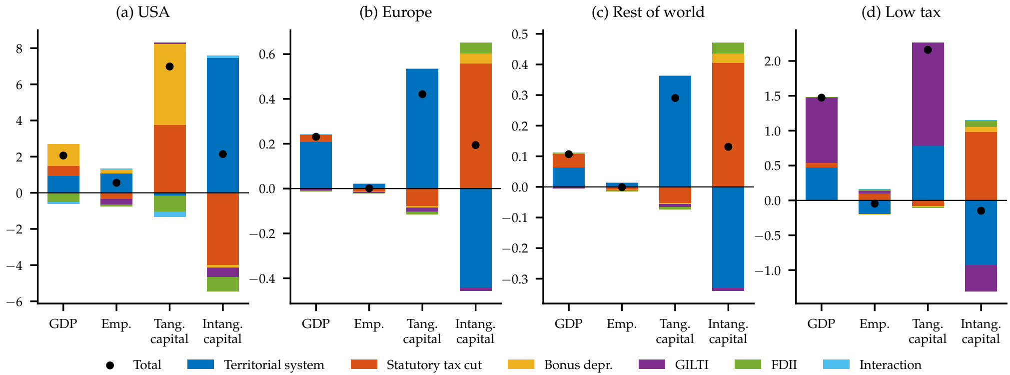

Territorial: intangible investment ↑ 7.46%; foreign profits are no longer taxed upon repatriation.

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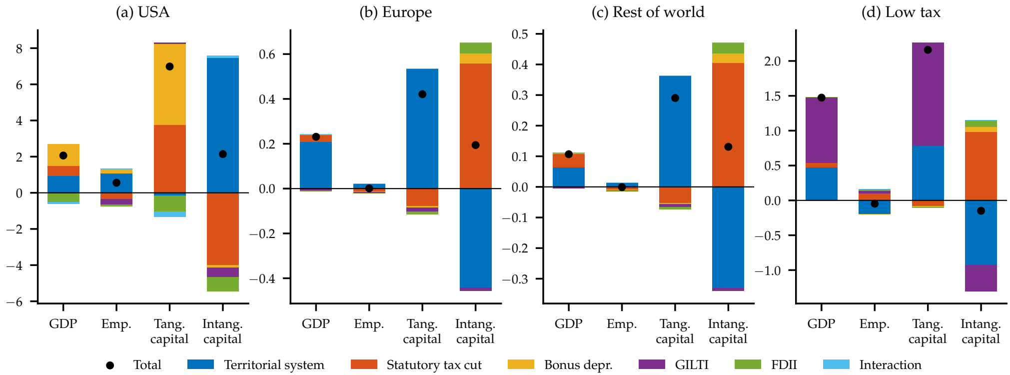

+ Rate cut (35% → 21%): benefits all U.S. firms; tangibles ↑, but intangibles ↓ (smaller PS gains).

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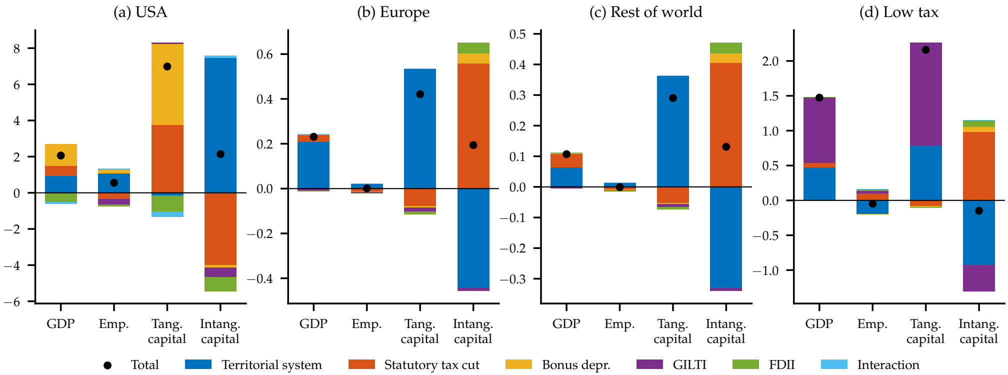

+ Bonus depreciation (50% → 100% expensing): the single largest GDP lever (+1.19 pp.).

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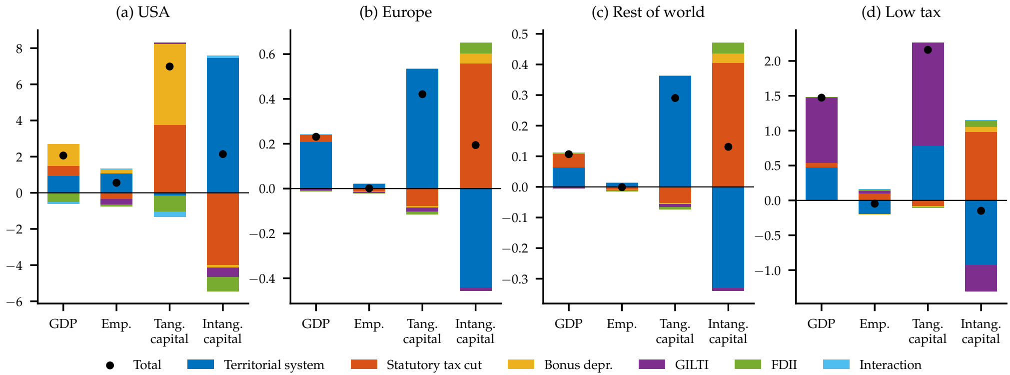
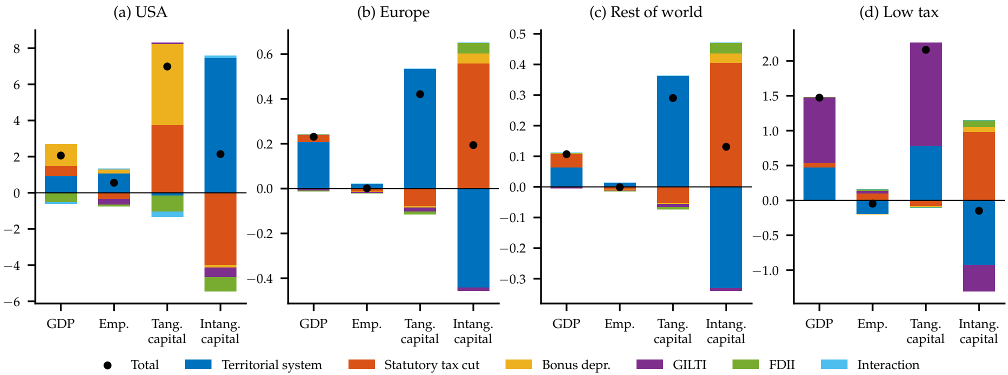


+ GILTI: near-zero GDP effect; its work is on profit shifting (next slide).

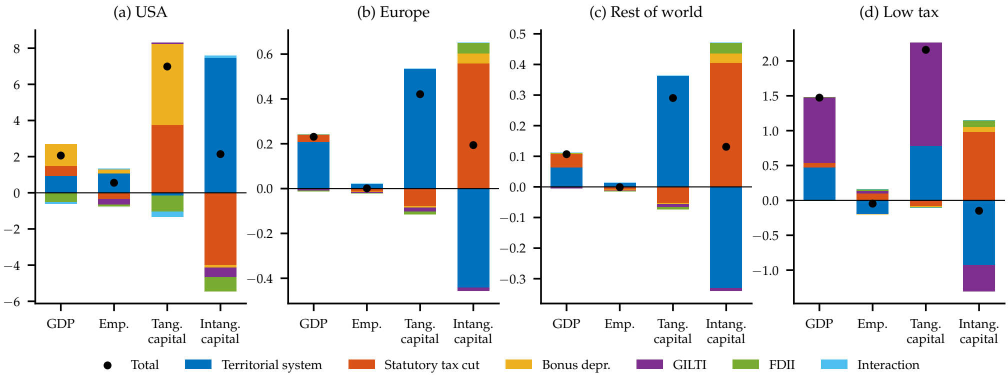
Global effects of TCJA: GDP and investment details by firm type incentives



+ FDII: ↓ tangibles via the QBAI penalty on domestic capital; GDP −0.5 pp.

Global effects of TCJA: GDP and investment

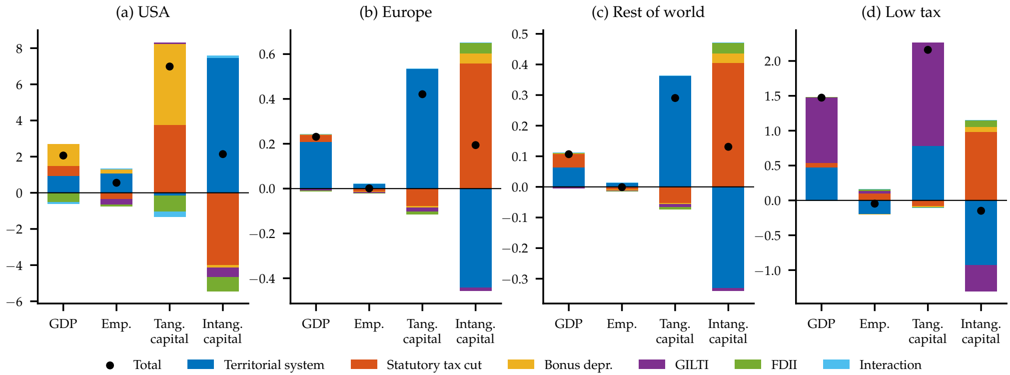
details | by firm type | incentives



Full TCJA: GDP ↑ 2.06%, tangible ↑ 6.98%, intangible ↑ 2.14%. Interactions are small ($\leq$ 0.3 pp.).

Global effects of TCJA: GDP and investment

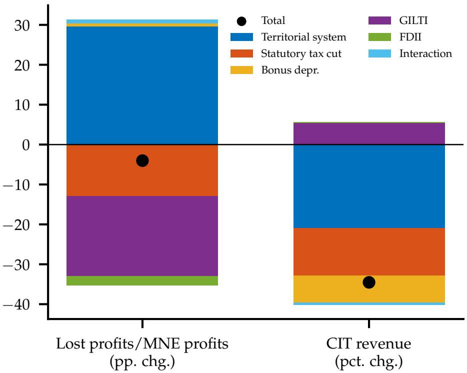
details by firm type incentives



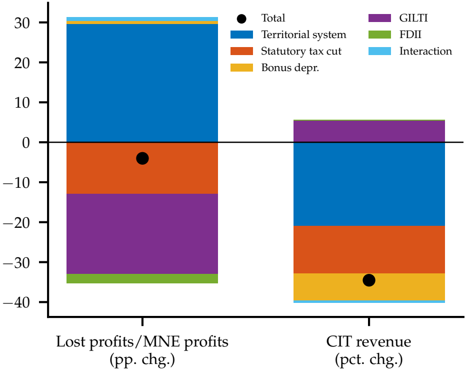
Positive ripple effects abroad: GDP ↑ 0.23% in Europe, ↑ 0.11% in the RoW.

1. U.S. MNEs ↑ nonrival intangibles at home ⇒ their foreign subsidiaries expand
2. foreign MNEs ↑ U.S. tangibles ⇒ ↑ intangibles at their HQs

TCJA reduces lost profits and CIT revenue details event study

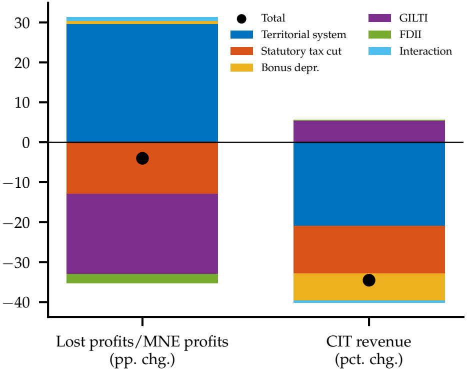


TCJA reduces lost profits and CIT revenue details event study



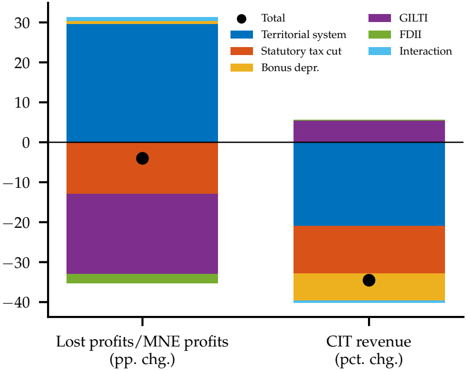
Territorial: shifting ↑ sharply (the repatriation friction is gone) and CIT revenue ↓ 21%.

TCJA reduces lost profits and CIT revenue `details` `event study`



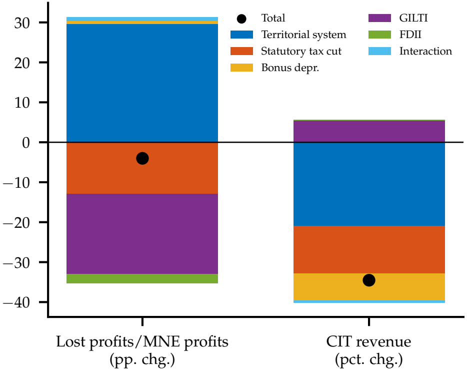
+ Rate cut: smaller tax differential ⇒ shifting ↓; revenue falls further.

TCJA reduces lost profits and CIT revenue details event study



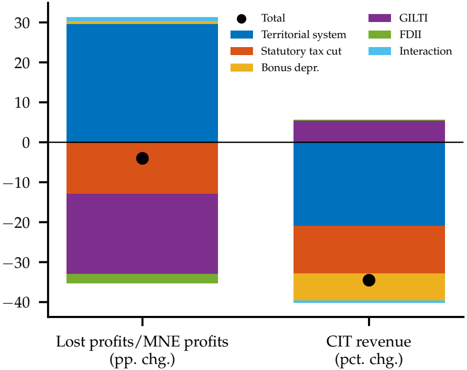
+ Bonus depreciation: no profit-shifting margin at all: a pure revenue cost (−6.7%).

TCJA reduces lost profits and CIT revenue details event study



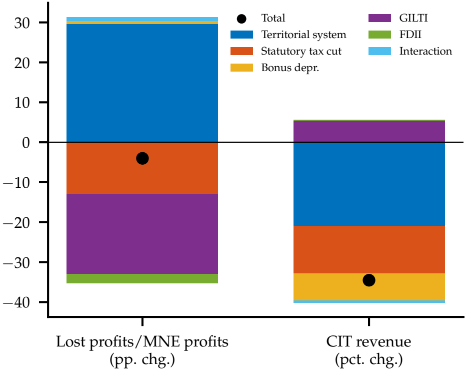
+ GILTI: the anti-shifting workhorse: −20 pp. of shifted profits, and the only provision that *materially* raises revenue.

TCJA reduces lost profits and CIT revenue details event study



+ FDII: the carrot: a small further decline in shifting.

TCJA reduces lost profits and CIT revenue details event study



Net effect: shifted-profit share ↓ 4.0 pp.; CIT revenue ↓ 34.5%.

Untargeted: share of shifted profits falls from 8.6% to 4.6% in the model vs. 7.6% to 5.1% in the data (Delis et al. 2024).

Validation I: the TCJA and firm-level investment `design` `GILTI/FDII exposure`

Event study on Compustat firms (2012–22). Exposure: a simulated tax change (as in Zwick and Mahon 2017)

$$\Delta\tau_i^{sim} = \frac{T^{post}(X_{i,2016}) - T^{pre}_{i,2016}}{\pi_{i,2016}}$$

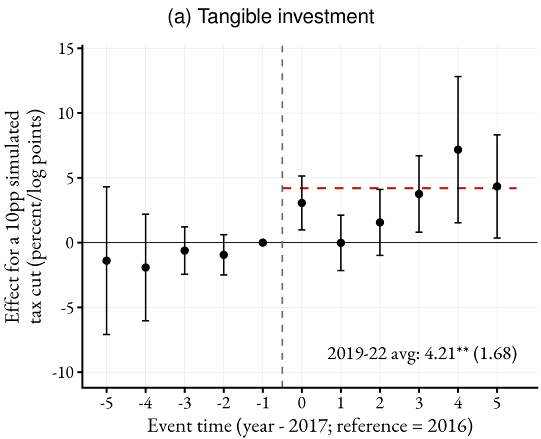
- post-TCJA rules applied to pre-reform (2016) firm characteristics: isolates the legislated change from firms' endogenous responses

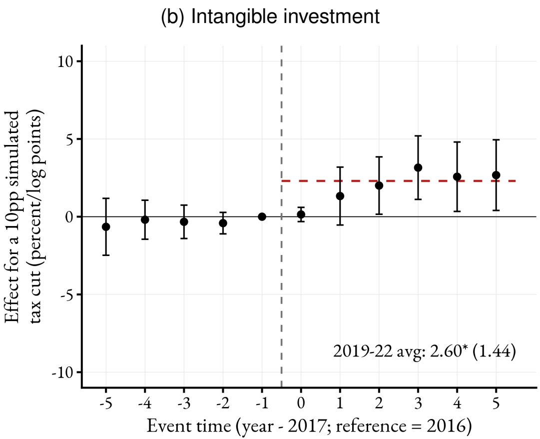
We estimate:

$$y_{it} = \alpha_i + \gamma_{n(i)\,m(i)\,t} + \sum_{\substack{\tau=-5\\ \tau\neq -1}}^{5} \beta_\tau\, \Delta\tau_i^{sim} \times \mathbf{1}\{t-2017=\tau\} + \varepsilon_{it}$$

- y_{it}: log tangible or log intangible investment
- α_i: firm FE; $\gamma_{n(i)m(i)t}$: industry $\times$ MNE status $\times$ year FE
- weighted by pre-reform revenue; SEs clustered at the firm level

Validation I: the TCJA and firm-level investment `design` `GILTI/FDII exposure`



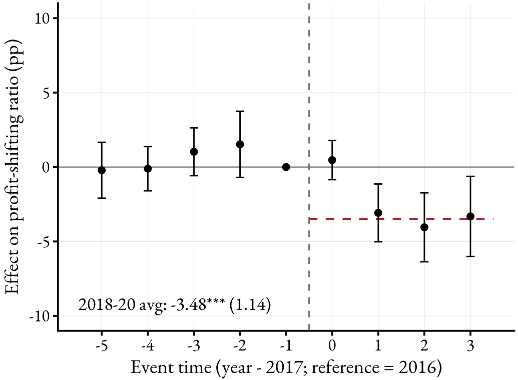


The mean simulated tax cut (10.2 pp.) ⇒ tangible investment ↑ 4.2%, intangible ↑ 2.6% (2019–22 avg.)

Flat pre-trends; tangible responds on impact, intangible builds gradually, consistent with costlier adjustment of intangibles.

Validation II: profit shifting and the model's fit details

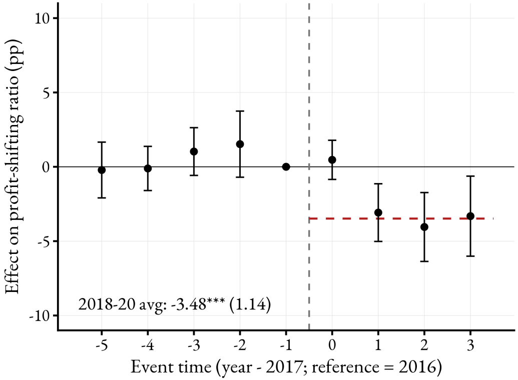
Shifted profits / total profits: U.S. vs. foreign MNEs



Shifted profits from Delis et al. (2024); design as in Crawford and Markarian (2024): U.S. vs. foreign MNEs in the U.S.

Validation II: profit shifting and the model's fit details

Shifted profits / total profits: U.S. vs. foreign MNEs



Shifted profits from Delis et al. (2024); design as in Crawford and Markarian (2024): U.S. vs. foreign MNEs in the U.S.

Model vs. data: the TCJA's effects

	Model	Data
Δ tangible capital (%)	$+7.0$	$+6.4$
		Chodorow-Reich et al. (2026), admin. data
Δ shifted-profit share (pp.)	-4.0	-3.5
		event study, left panel
Shifted share, pre $\rightarrow$ post (%)	$8.6 \rightarrow 4.6$	$7.6 \rightarrow 5.1$
		untargeted levels; Delis et al. (2024)

Counterfactual experiments: GMT

Global effects of the GMT (relative to the TCJA equilibrium) by firm type QBAI vs SBIE

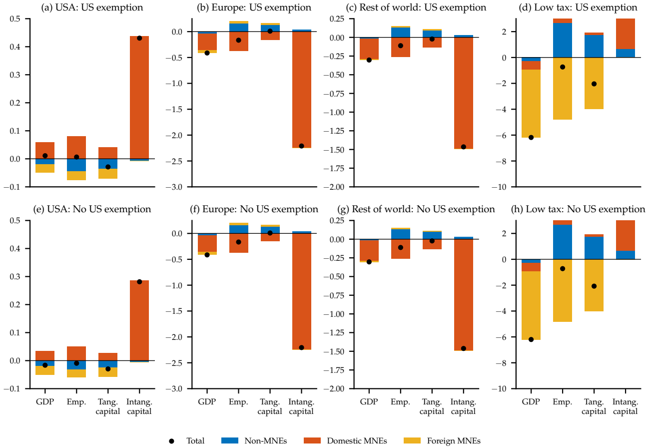
GMT at 15% adopted by Europe, RoW, Low-Tax, and Tax Haven; not by the U.S. Two regimes for U.S. MNEs:

1. Complete exemption: the GMT applies only to non-U.S. MNEs
2. Side-by-Side: no IIR/UTPR on U.S. MNEs and GILTI stays intact, but LT and TH levy QDMTT top-ups on U.S. MNEs' local profits
   - $\rightarrow$ top-up rate $=$ 15% $-$ effective GILTI rate: 13.8% (LT) and 12.2% (TH) $\Rightarrow$ top-ups of 1.2 and 2.8 pp.
   - $\rightarrow$ the SBIE carve-out ($5\%K + 5\%wL \approx 6\%$ of K) is less generous than the QBAI (10% of K)

Both computed from the post-TCJA steady state, holding U.S. policy fixed.

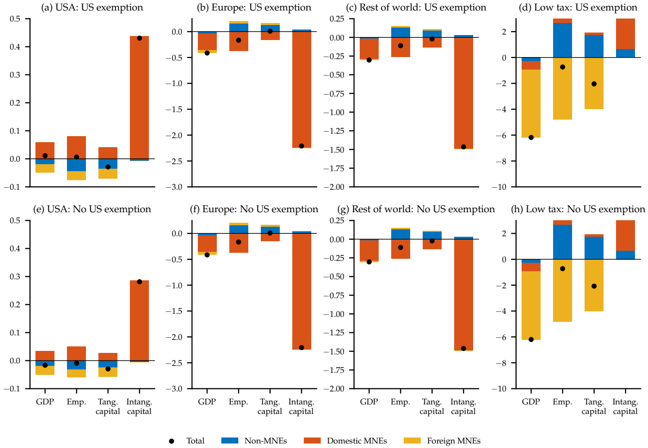
Note: quantitative results retain the original TCJA GILTI/FDII parameters; the OBBBA reparameterization is not modeled.

Global effects of the GMT (relative to the TCJA equilibrium) by firm type QBAI vs SBIE



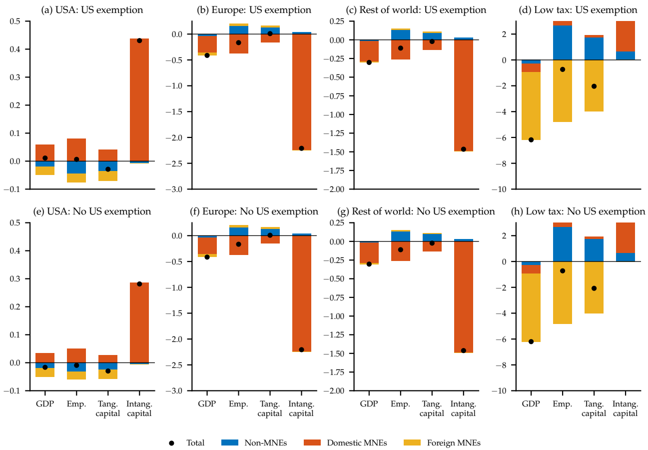
Adopting high-tax regions: intangibles ↓ 2.2% (Europe) and 1.5% (RoW) ⇒ GDP ↓ 0.4% and 0.3%.

Global effects of the GMT (relative to the TCJA equilibrium) by firm type QBAI vs SBIE



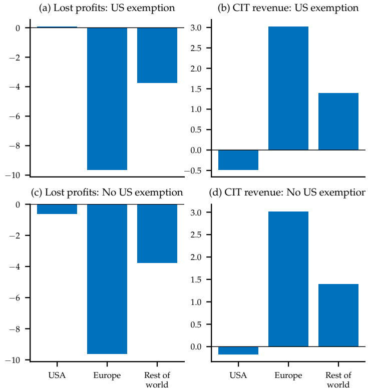
Low-tax region: GDP ↓ 6.2%, but intangibles ↑ 5.0%: domestic firms replace departing MNEs' nonrival z.

Global effects of the GMT (relative to the TCJA equilibrium) by firm type QBAI vs SBIE

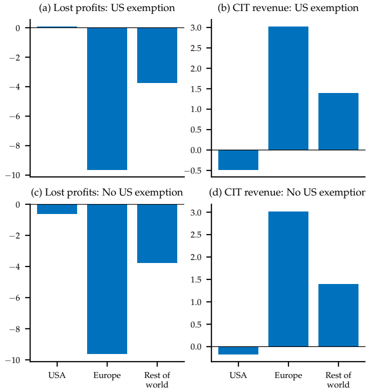


U.S.: GDP +0.01% if exempt vs. −0.02% under Side-by-Side; the sign of the inward ripple is a treaty detail.

The GMT: profit shifting and public finances `by firm type`

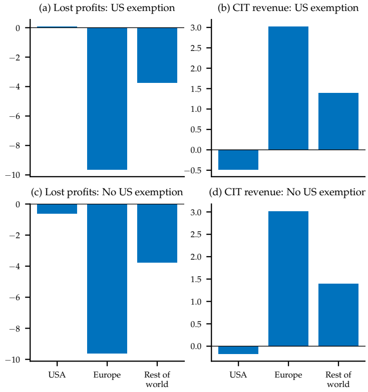


The GMT: profit shifting and public finances `by firm type`



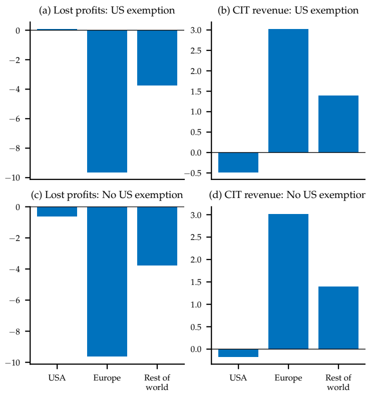
Shifting collapses for European (−9.6 pp.) and RoW (−3.8 pp.) MNEs; their CIT revenues rise 3.0% and 1.4%.

The GMT: profit shifting and public finances by firm type



Rule priority means the low-tax region wins: QDMTT top-ups more than replace the lost shifting inflows (CIT ↑ 6%).

The GMT: profit shifting and public finances `by firm type`



U.S. CIT falls either way: more QBAI deductions if exempt; top-ups paid abroad under Side-by-Side, which also cuts U.S. shifting by 0.6 pp. more than GILTI alone.

Conclusion

Conclusions AETR/METR

- TCJA: a large domestic expansion at a fiscal cost, with positive ripple effects abroad

- GMT: revenue up and activity down where it binds; the U.S. inward ripple is decided by the Side-by-Side fine print, so institutional specifics, not headline rates, determine the magnitudes

- Per tax point moved, the GMT's cross-border effects rival the TCJA's (the U.S. effect is NAFTA-sized; Caliendo and Parro 2014):

  there is no such thing as a purely domestic corporate tax policy

Additional Slides

Contributions to the literature

1. **Profit shifting:** Hines and Rice (1994), Heckemeyer and Overesch (2017), Suárez-Serrato (2018), Beer, de Mooij, and Liu (2020), Clausing (2020), Guvenen et al. (2022), Tørsløv et al. (2023), Ferrari et al. (2023), Bilicka et al. (2024), Delis et al. (2024), Dyrda et al. (2024, 2025), Altshuler et al. (2025)

   → Evaluate the global effects of regional tax reforms using a multi-country GE model

2. **TCJA and GMT:** Auerbach (2018), Barro and Furman (2018), Beer et al. (2018), Slemrod (2018), Gale et al. (2019), Wagner et al. (2020), Johannesen (2022), Garcia-Bernardo et al. (2023), Clausing (2024), Hugger et al. (2024), Santacreu and Stewart (2024), Chodorow-Reich et al. (2025), Dobridge et al. (2025)

   → Model the tax provisions for MNEs and study the impacts on both tangible and intangible investment

3. **Multinational enterprises:** Helpman et al. (2004), Burstein and Monge-Naranjo (2009), McGrattan and Prescott (2009), Ramondo and Rodríguez-Clare (2013), Tintelnot (2017), Arkolakis et al. (2018), McGrattan and Waddle (2020), Alviarez et al. (2023), Antràs et al. (2025)

   → Emphasize the role of MNE networks and intangibles in the ripple effects of regional tax reforms

The Model

Technology spillovers return

With technology spillovers, the production function of intangible capital is:

$$z_t(\omega) = A_i \ell_{it}^z(\omega) \times \underbrace{\left[\sum_{j \neq i} \int_{\Omega_{jit}} z_{jt}(\omega') \, \mathrm{d}\omega' \right]^{\upsilon}}_{\text{Intangibles used by foreign MNEs}}$$

– where υ is the technology spillover elasticity; $\upsilon = 0.4$ following Dyrda, Hong and Steinberg

Households: preferences and budgets

- In each region i representative household solves:

$$\max_{\{C_{it},L_{it},X_{it},B_{it+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left[\log\left(\frac{C_{it}}{N_i}\right) + \psi_i \log\left(1 - \frac{L_{it}}{N_i}\right)\right].$$

  where C_{it} is consumption, L_{it} is labor supply.
- Budget constraint:

$$P_{it}[(1+\tau_{ic})C_{it} + X_{it}] + P_{bt}B_{it+1} = (1-\tau_{i\ell})W_{it}L_{it} + R_{it}K_{it} + B_{it} + D_{it},$$

  where X_{it} is tangible investment, B_{it+1} are internationally-traded bonds, D_{it} dividends of MNEs headquartered in i, and $\tau_{ic}, \tau_{i\ell}$ are consumption and labor income tax rates.

  The law of motion for tangible capital:

$$K_{it+1} = (1-\delta)K_{it} + X_{it},$$

return

Final Goods Producers

In each region i representative final-good producer that combines domestic and foreign products into a nontradable aggregate:

$$Q_{it} = \left[\sum_{j=1}^{I} \int_{\Omega_{jit}} q_{jit}(\omega)^{\frac{\varrho-1}{\varrho}} d\omega \right]^{\frac{\varrho}{\varrho-1}},$$

where $q_{jit}(\omega)$ is the quantity of variety ω from region j, Ω_{jit} is the set of goods from j available in i (determined by firms' exporting and FDI decisions specified later).

The aggregate price index is:

$$P_{it} = \left[\sum_{j=1}^{I} \int_{\Omega_{jit}} p_{jit}(\omega)^{1-\varrho} d\omega \right]^{\frac{1}{1-\varrho}}$$

return

Aggregation and accounting measures: GDP and Goods trade

- Gross domestic product:

$$GDP_i = \sum_{j=1}^{I} \int_{\omega \in \Omega_j, i \in J_F(\omega)} p_{ji}(\omega) y_{ji}(\omega) \, d\omega.$$

- Goods trade:

$$EX_i^G = \sum_{j \neq i} \int_{\Omega_i} p_{ij}^X(\omega) \, \xi_{ij} \, q_{ij}^X(\omega) \, d\omega,$$

$$IM_i^G = \sum_{j \neq i} \int_{\Omega_j} p_{ji}^X(\omega) \, \xi_{ji} \, q_{ji}^X(\omega) \, d\omega.$$

Scale choice return

- The domestic parent corporation's problem (omit firm identifier ω):

$$\hat{\pi}_{ii}(a,z,J_X) = \max_{q_{ii},\{q_{ij}^X\}_{j\in J_X},\ell_{ii},k_{ii}} (1-\tau_i)\underbrace{\Big(p_{ii}(q_{ii})q_{ii} + \sum_{j\in J_X} p_{ij}(q_{ij}^X)q_{ij}^X - W_i\ell_{ii} - \delta P_i k_{ii}\Big)}_{:=\pi_{ii}^D(a,z)} - (1-\tau_i b_i)\, r_i k_{ii}$$

s.t $\quad q_{ii} + \sum_{j\in J_X} \xi_{ij} q_{ij}^X = y_{ii}.$

$\rightarrow$ the parent corporation can both sell to the home country and export to foreign countries

Scale choice return

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- Foreign subsidiaries' problem:

$$\hat{\pi}_{ij}(a,z) = \max_{q_{ij},\ell_{ij},k_{ij}} (1-\tau_j)\Big(\underbrace{p_{ij}(q_{ij})q_{ij} - W_j\ell_{ij} - \delta P_j k_{ij}}_{:=\pi_{ij}^F(a,z)}\Big) - (1-\tau_j b_j)\, r_j k_{ij},\ j\in J_F.$$

  $\rightarrow$ the foreign subsidiaries can only sell to the local households, assume that the foreign subsidiary produces another variety than the parent.

  $\rightarrow$ bonus depreciation: only the share $1-b_j$ of the net rental cost $r_j k$ remains non-deductible ($b_j = 0$ outside the U.S.)

Scope choice return

- MNE maximizes after-tax dividends:

$$d_i(a) = \max_{J_X, J_F, z, \lambda_{LT}, \lambda_{TH}} \left\{ \pi_{ii}(a, z, J_X) - T_{ii} + \sum_{j \in J_F} (\pi_{ij} - T_{ij}) + \mathbb{1}_{\{\lambda_{TH}>0\}} \pi_{i,TH} - T_{i,TH} \right\}$$

subject to

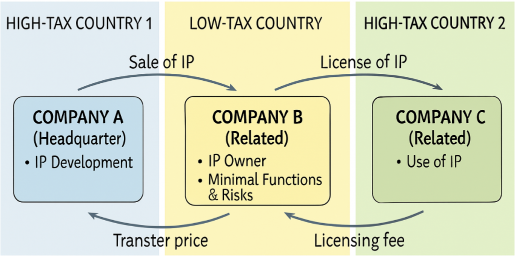
$$\Gamma = \left\{ \boldsymbol{\lambda} \in [0,1]^2 : \lambda_{LT} + \lambda_{TH} \leq 1 \right\}.$$

where:

- $\rightarrow$ π_{ii}: pre-tax profits of the parent division
- $\rightarrow$ π_{ij}: pre-tax profits of affiliates in region j
- $\rightarrow$ $\pi_{i,LT}$: pre-tax profits of the low-tax affiliate
- $\rightarrow$ $\pi_{i,TH}$: pre-tax profits of the tax-haven affiliate
- $\rightarrow$ T_{ij}: tax liabilities paid in jurisdiction j (levied on taxable income $\tilde{\pi}_{ij}$)
- $\rightarrow$ $\boldsymbol{\lambda} = (\lambda_{LT}, \lambda_{TH})$: shares of rights to intangible capital sold to LT and TH

Profit shifting via transferring intangible capital return

- MNEs shift profits by transferring nonrival intangible capital to affiliates in tax havens
- Tax-haven affiliates charge parent (and other affiliates) licensing fees



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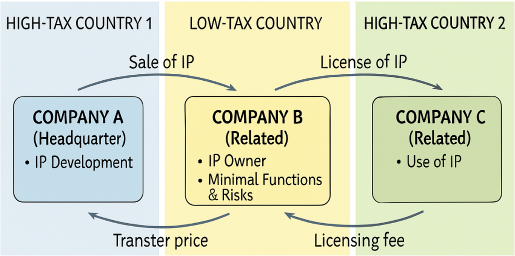

"95 percent of Apple's R&D... is conducted in the United States... [During] 2009 to 2012, ASI [Apple Ireland] paid... $5 billion to [Apple USA] as its share of the R&D costs. Over that same time period, ASI received profits of $74 billion. The difference between ASI's costs and the profits, almost $70 billion, is how much taxable income [should] have flowed to the United States." (Levin and McCain 2013, p. 2)

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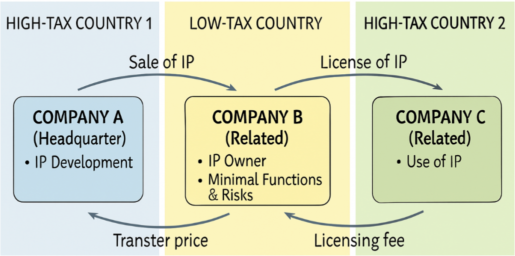

- End result: ↓ global tax liability and ↑ after-tax return on intangible investment
- Both TCJA and GMT are designed to decrease the returns of transferring intangibles

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Pre-tax profits: parent division return

$$\pi_{ii} = p_{ii}(q_{ii})q_{ii} + \sum_{j \in J_X} p_{ij}(q_{ij})q_{ij} - W_i \overbrace{\left(l_{ii} + l_i^z + \sum_{j \in J_X} \kappa_{iX} + \sum_{j \in J_F} \kappa_{iF} + \kappa_{i,TH} \mathbb{1}_{\{\lambda_{TH} > 0\}} \right)}^{\text{Labor cost: production labor, intangible investment and fixed costs}}$$

$$+ \overbrace{(\varphi_{iLT}\lambda_{LT} + \varphi_{iTH}\lambda_{TH})\,\nu_i(z)z}^{\text{Proceeds from selling } z} + \overbrace{\sum_{j \in J_F} (1 - \lambda_{LT} - \lambda_{TH})\vartheta_{ij}(z)z}^{\text{Licensing fee receipts}} - \overbrace{(\lambda_{LT} + \lambda_{TH})\vartheta_{ii}(z)z}^{\text{Licensing fee payments}}$$

$$- \overbrace{W_i \left[\mathcal{C}_{i,LT}(\lambda_{LT}) + \mathcal{C}_{i,TH}(\lambda_{TH})\right] \nu_i(z)z}^{\text{Cost of transferring } z} - \overbrace{\delta P_i k_{ii}}^{\text{Depreciation}} - \overbrace{r_i k_{ii}}^{\text{Net rental cost}}$$

where:

- $\rightarrow$ κ_{iX}: a fixed cost to export domestically produced goods
- $\rightarrow$ κ_{iF}: a fixed cost to open a foreign affiliate and produce locally; $\kappa_{i,TH}$: fixed cost of the tax-haven affiliate (paid iff $\lambda_{TH} > 0$)
- $\rightarrow$ $\vartheta_{ij}(z)$: the firm's marginal revenue product of intangible capital in j; $\vartheta_{ij}(z)z$ is the licensing fee paid by the division in j
- $\rightarrow$ $\nu_i(z)z \equiv \sum_{j \in J_F \cup \{i\}} \vartheta_{ij}(z)z$: total amount of licensing fees across the conglomerate
- $\rightarrow$ $\varphi_{iLT}, \varphi_{iTH}$: markdowns (mispricing) on selling rights to intangible capital

Pre-tax profits: foreign subsidiaries return

- Foreign subsidiary in high-tax region $j \notin \{LT, TH\}$:

$$\pi_{ij} = p_{ij}(\hat{q}_{ij})\hat{q}_{ij} - W_j \ell_{ij} - \delta P_j k_{ij} - \underbrace{\vartheta_{ij}(z)z}_{\text{Licensing fee payments}} - r_j k_{ij}$$

Pre-tax profits: foreign subsidiaries return

- Foreign subsidiary in high-tax region $j \notin \{LT, TH\}$:

$$\pi_{ij} = p_{ij}(\hat{q}_{ij})\hat{q}_{ij} - W_j \ell_{ij} - \delta P_j k_{ij} - \underbrace{\vartheta_{ij}(z)z}_{\text{Licensing fee payments}} - r_j k_{ij}$$

- Low Tax (LT):

$$\pi_{i,LT} = p_{i,LT}(\hat{q}_{i,LT})\hat{q}_{i,LT} - W_{LT}\ell_{i,LT} - \delta P_{LT} k_{i,LT} - r_{LT} k_{i,LT} - \overbrace{(1-\lambda_{LT})\vartheta_{i,LT}(z)z}^{\text{Licensing fee payments}}$$
$$+ \underbrace{\sum_{j \in J_F \cup \{i\} \setminus \{LT\}} \lambda_{LT}\vartheta_{ij}(z)z}_{\text{Licensing fee receipts}} - \underbrace{\varphi_{iLT}\lambda_{LT}\nu_i(z)z}_{\text{Cost of buying } z}$$

Pre-tax profits: foreign subsidiaries return

- Foreign subsidiary in high-tax region $j \notin \{LT, TH\}$:

$$\pi_{ij} = p_{ij}(\hat{q}_{ij})\hat{q}_{ij} - W_j \ell_{ij} - \delta P_j k_{ij} - \underbrace{\vartheta_{ij}(z)z}_{\text{Licensing fee payments}} - r_j k_{ij}$$

- Low Tax (LT):

$$\pi_{i,LT} = p_{i,LT}(\hat{q}_{i,LT})\hat{q}_{i,LT} - W_{LT}\ell_{i,LT} - \delta P_{LT} k_{i,LT} - r_{LT} k_{i,LT} - \overbrace{(1-\lambda_{LT})\vartheta_{i,LT}(z)z}^{\text{Licensing fee payments}}$$
$$+ \underbrace{\sum_{j \in J_F \cup \{i\} \setminus \{LT\}} \lambda_{LT}\vartheta_{ij}(z)z}_{\text{Licensing fee receipts}} - \underbrace{\varphi_{iLT}\lambda_{LT}\nu_i(z)z}_{\text{Cost of buying } z}$$

- Tax Haven (TH):

$$\pi_{i,TH} = \underbrace{\sum_{j \in J_F \cup \{i\}} \lambda_{TH}\vartheta_{ij}(z)z}_{\text{Licensing fee receipts}} - \underbrace{\varphi_{iTH}\lambda_{TH}\nu_i(z)z}_{\text{Cost of buying } z}$$

Aggregation and accounting measures: Services trade

- High-tax regions' services:

$$EX_i^S = \sum_{j \neq i} \int_{\Omega_i} \left[1 - \lambda_{LT}(\omega) - \lambda_{TH}(\omega)\right] \vartheta_{ij}(\omega) z(\omega)\, d\omega + \int_{\Omega_i} \left[\varphi_{i,LT} \lambda_{LT}(\omega) + \varphi_{i,TH} \lambda_{TH}(\omega)\right] \nu_i(\omega) z(\omega)\, d\omega$$

$$IM_i^S = \sum_{j \neq i} \int_{\Omega_i} \left[\lambda_{LT}(\omega) + \lambda_{TH}(\omega)\right] \vartheta_{ij}(\omega) z(\omega)\, d\omega + \sum_{j \neq i} \int_{\Omega_j} \vartheta_{ji}(\omega) z(\omega)\, d\omega.$$

- The low-tax region's services:

$$\begin{aligned} EX_{LT}^S &= \sum_{j \neq LT} \int_{\Omega_{LT}} \left[1 - \lambda_{TH}(\omega)\right] \vartheta_{LT,j}(\omega) z(\omega)\, d\omega + \sum_{j \neq LT} \int_{\Omega_j} \lambda_{LT}(\omega) \vartheta_{j,LT}(\omega) z(\omega)\, d\omega, \\ IM_{LT}^S &= \sum_{j \neq LT} \int_{\Omega_{LT}} \lambda_{TH}(\omega) \vartheta_{LT,j}(\omega) z(\omega)\, d\omega + \sum_{j \neq LT} \int_{\Omega_j} [1 - \lambda_{LT}(\omega)] \vartheta_{j,LT}(\omega) z(\omega)\, d\omega + \\ &\quad \sum_{j \neq LT} \int_{\Omega_j} \varphi_{j,LT} \lambda_{LT}(\omega) \nu_j(\omega) z(\omega)\, d\omega. \end{aligned}$$

Market clearings

- Labor market:

$$L_i = \overbrace{\sum_{j=1}^{I} \int_{\Omega_j} \ell_{ji}(\omega)\, d\omega}^{\text{goods production}} + \overbrace{\int_{\Omega_i} l_i^z \, d\omega}^{z \text{ production}} + \overbrace{\int_{\Omega_i} \left(\sum_{j \in J_X(\omega)} \kappa_{iX} + \sum_{j \in J_F(\omega)} \kappa_{iF} + \mathbb{1}_{\{\lambda_{TH}(\omega)>0\}} \kappa_{i,TH} \right) d\omega}^{\text{fixed costs}}$$
$$\underbrace{+ \int_{\Omega_i} \left[\mathcal{C}_{i,LT}(\lambda_{LT}) + \mathcal{C}_{i,TH}(\lambda_{TH}) \right] \nu(\omega) z(\omega)\, d\omega}_{\text{costs of shifting } z} .$$

- Capital market:

$$K_i = \sum_{j=1}^{I} \int_{\Omega_{ji}} k_{ji}(\omega)\, d\omega$$

- Government budget constraint (G_i held fixed in counterfactuals; T_{ji}: taxes an MNE from j pays in i):

$$P_i G_i = \tau_{i\ell} W_i L_i + \sum_{j=1}^{I} \int_{\Omega_j} T_{ji}(\omega)\, d\omega,$$

return

Market clearings

- Balance of payments:

$$EX_i^G + EX_i^S - IM_i^G - IM_i^S + NFR_i - NFP_i = P_b B_i' - B_i \ (= 0 \text{ in stationary eq.}).$$

where:

$$NFR_i = \sum_{j \neq i} \int_{\Omega_{ij}} \left[\pi_{ij}(\omega) - T_{ij}(\omega)\right] \, d\omega,$$
$$NFP_i = \sum_{j \neq i} \int_{\Omega_{ji}} \left[\pi_{ji}(\omega) - T_{ji}(\omega)\right] \, d\omega.$$

are net factor receipts from (payments to) foreigners.

return

Tax reforms: TCJA and GMT

U.S. taxation before the TCJA return

- Prior to the TCJA, the U.S. operated a worldwide taxation system:
  - $\rightarrow$ Foreign income of U.S.-based MNEs was subject to U.S. taxation of 35% upon repatriation.
  - $\rightarrow$ Firms claim a 100% foreign tax credit for taxes paid to foreign gvnt, subject to limitations.
- We assume that U.S.-based multinational enterprises (MNEs) repatriate a share ι_j of income from region j.
- The domestic tax liability of a U.S. (i) parent is given by:

$$T_{ii} = \tau_i^{pre}\tilde{\pi}_{ii} + \sum_{j \in J_F \cup \{TH\}} \iota_j \max\left(0, (\tau_i^{pre} - \tau_j)\tilde{\pi}_{ij}\right)$$

$\tilde{\pi}_{ij} = \pi_{ij} + (1 - b_j)r_j k_{ij}$: taxable income under bonus depreciation bonus

U.S. taxation after the TCJA (I) return

- Global Intangible Low-Taxed Income (GILTI) is a minimum tax on foreign income over the routine return:

$$T_{ii}^{GILTI} = \max\left\{0,\ \tau_i(1-\chi^{GILTI}) \sum_{j\in J_F\setminus J_F^{HT}} \left(\tilde{\pi}_{ij} - \chi^{QBAI} P_j k_{ij}\right) - T_{ii}^{FTC}\right\}$$

→ $\chi^{GILTI} = 0.5$: the Section 250 deduction rate

→ $\chi^{QBAI} = 0.1$: the deemed routine return on tangible assets

→ J_F^{HT}: the set of high-tax jurisdictions whose ETR are at least 90% of U.S. tax rate

→ T_{ii}^{FTC}: the allowable foreign tax credit ($\chi^{FTC} = 0.8$): the minimum of deemed-paid taxes and the U.S. tax on the limitation base; in our implementation the deemed-paid term binds

$$T_{ii}^{FTC} = \min\left(\chi^{FTC} \sum_{j\in J_F\setminus J_F^{HT}} \tau_j \tilde{\pi}_{ij},\ \tau_i \times \max\left(0, (1-\chi^{GILTI})\tilde{\pi}_{ii}^{GILTI} - \sum_{j\in J_F\setminus J_F^{HT}} \kappa_{iF}\right)\right)$$

US taxation after the TCJA (II) return

- Foreign Derived Intangible Income (FDII) is an subsidy on export income over the routine return:

$$T_{ii}^{FDII} = \tau_i \times \chi^{FDII} \times \left(\tilde{\pi}_{ii}^{FDDEI} - \chi^{QBAI} \times p_i k_{ii} \times \underbrace{\frac{\tilde{\pi}_{ii}^{FDDEI}}{\tilde{\pi}_{ii}}}_{FDR} \right)$$

  → $\chi^{FDII} = 0.375$: the deduction rate applied to eligible income of FDII
  → $\tilde{\pi}_{ii}^{FDDEI}$: the foreign-derived deduction-eligible income: gross foreign receipts − allocated deductions
  → $\tilde{\pi}_{ii}$: total taxable income; the ratio is the foreign-derived ratio (FDR), fixed at its pre-TCJA value

- The domestic tax liability of a U.S. parent is given by:

$$T_{ii} = \tau_i \tilde{\pi}_{ii} + T_{ii}^{GILTI} - T_{ii}^{FDII}$$

Section 250: deduction limitation of GILTI and FDII return

- The deduction for FDII and GILTI is limited if the sum of these inclusions exceeds the corporation's taxable income, i.e. $\tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii} > 0$

$$R_{ii}^{FDII} = \begin{cases} 0 & \text{if } \tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii} \leq 0 \\ \frac{\tilde{\pi}_{ii}^{FDII}}{\tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII}} \times \left(\tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii}\right) & \text{if } \tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii} > 0 \end{cases}$$

$$R_{ii}^{GILTI} = \begin{cases} 0 & \text{if } \tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii} \leq 0 \\ \frac{\tilde{\pi}_{ii}^{GILTI}}{\tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII}} \times \left(\tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii}\right) & \text{if } \tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} - \tilde{\pi}_{ii} > 0 \end{cases}$$

- Then

$$D^{FDII} = \chi^{FDII} \times \left(\tilde{\pi}_{ii}^{FDII} - R^{FDII}\right), \qquad D^{GILTI} = \chi^{GILTI} \times \left(\tilde{\pi}_{ii}^{GILTI} - R^{GILTI}\right)$$

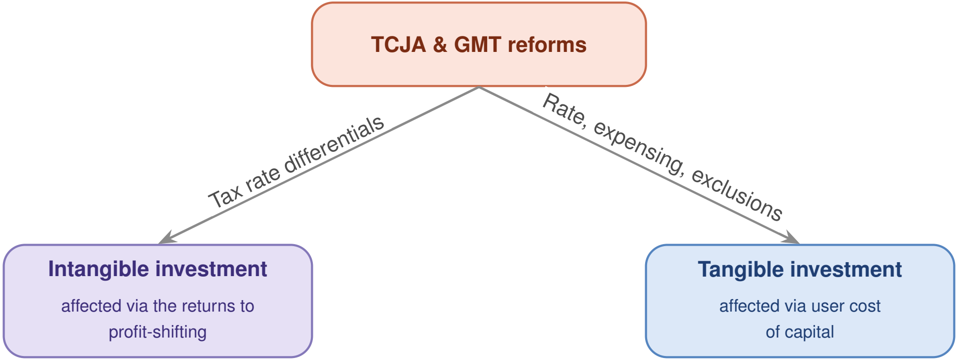
- In our implementation $\tilde{\pi}_{ii}^{GILTI} + \tilde{\pi}_{ii}^{FDII} < \tilde{\pi}_{ii}$, so $R^{FDII} = R^{GILTI} = 0$ (the limitation does not bind).
- The total domestic tax liability of a U.S. parent is

$$T_{ii} = \tau_i \left(\tilde{\pi}_{ii}^{DEI} + \tilde{\pi}_{ii}^{GILTI} - D^{FDII} - D^{GILTI}\right) - T_{ii}^{FTC}$$

The impacts of the tax reforms on investment

signs by provision | QBAI vs SBIE | return

The reforms affect firms' investment through tax rates, expensing, and base exclusions:



The impacts of the tax reforms on investment signs by provision QBAI vs SBIE return

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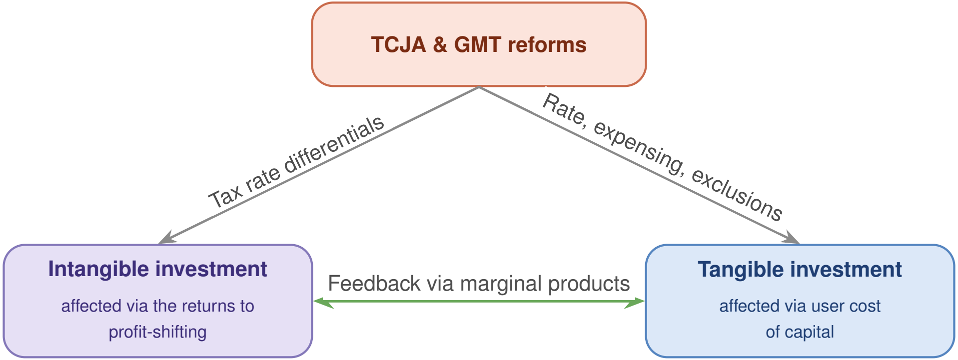

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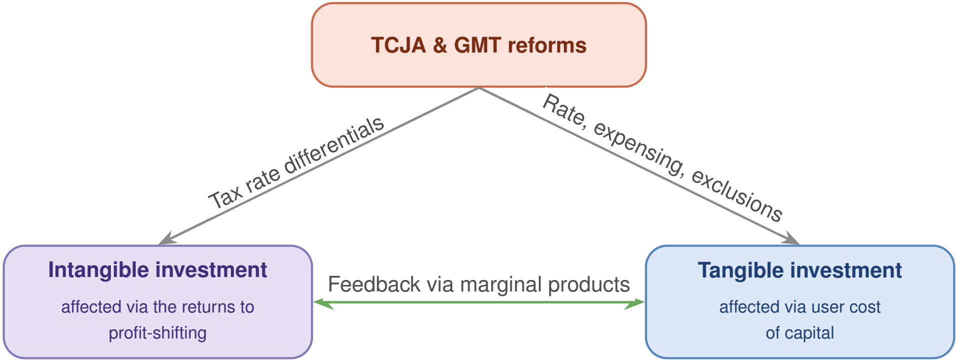


Bonus depreciation: a pure user-cost tool: no profit-shifting margin; substitutes with FDII's rate-cut channel

Directional effects of TCJA provisions and GMT on U.S. outcomes return

		Direct		Indirect	
Policy	**Profit shifting** λ	**Tangible inv.** k	**Intangible inv.** z	**Tangible inv.** k	**Intangible inv.** z
GILTI	$\downarrow$	–	$\downarrow$	?	$\uparrow$
FDII	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
GMT	$\downarrow$	–	$\downarrow$	?	$\uparrow$

Notes: All three provisions reduce the incentive to shift profits by compressing the U.S.–foreign tax wedge, which implies they all have a negative direct effect on intangibles. GILTI boosts tangible investment in low-tax affiliates through its QBAI exclusion, which has the indirect effect of increasing intangible investment. The indirect effect on domestic tangible investment can be positive or negative, depending on whether intangible investment ultimately rises or falls. FDII increases the effective tax rate on domestic tangibles, which directly lowers tangible investment and reinforces the direct effect on intangibles. GMT's substance-based carve-outs operate similarly to the GILTI QBAI exclusion, raising tangible investment in low-tax jurisdictions and thus indirectly raising intangible investment.

Other Provisions in TCJA return

- Base-Erosion and Anti-Abuse Tax (BEAT)
  - $\rightarrow$ Minimum tax on large MNEs making excessive deductible payments to foreign affiliates
  - $\rightarrow$ Applies to firms with gross receipts $\geq$ \$500M and base erosion payments $>$ 3% of deductions
  - $\rightarrow$ Largely avoided by decreasing base erosion rate; revenue far below projections
- Accelerated / Bonus Depreciation
  - $\rightarrow$ 100% immediate expensing of short-lived tangible assets
  - $\rightarrow$ Initially planned to phase out from 2023; made permanent by OBBBA
- Section 174: R&E Amortization, from 2022
  - $\rightarrow$ Replaced immediate R&E expensing with 5-year (domestic) or 15-year (foreign) amortization
  - $\rightarrow$ Partially reversed by the One Big Beautiful Bill Act (2025) for domestic R&E only
- Section 163(j): Interest Deduction Limitation
  - $\rightarrow$ Caps net business interest deductibility at 30% of adjusted taxable income
  - $\rightarrow$ Targets inter-company debt shifting; affected firms responded by increasing equity issuance

The OECD Pillar 2: Global Minimum Tax return

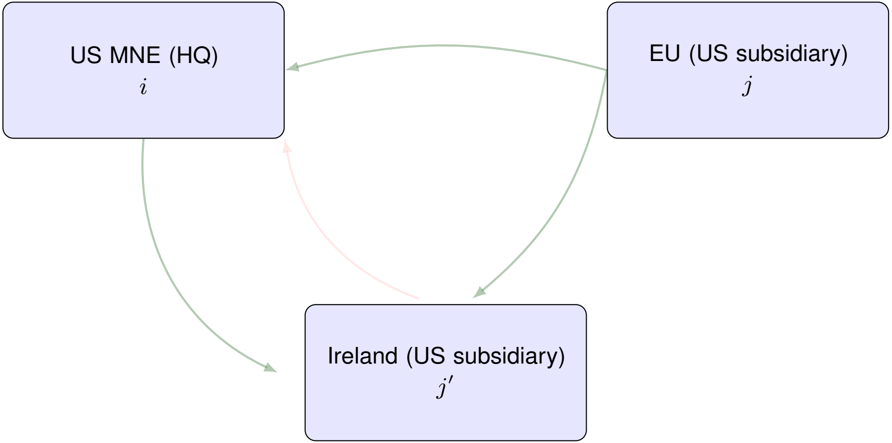
- The GMT tax liability is:

$$T_{ij}^{GMT} = \mathbb{1}\{j \in J^{QDMTT}\} \max\left\{0,\ \tau_{GMT} - \tau_{ij}^{eff}\right\} (\tilde{\pi}_{ij} - \chi_{GMT,K} P_j k_{ij} - \chi_{GMT,L} W_j \ell_{ij})$$

  - $\rightarrow$ $\tau_{GMT} = 15\%$: GMT rate
  - $\rightarrow$ τ_{ij}^{eff}: the effective rate of the region i's subsidiary in j
  - $\rightarrow$ $\chi_{GMT,K} = \chi_{GMT,L} = 5\%$: substance-based carve-out rates of tangible capital and payroll

- The taxing right of the top-up tax is based on the three-tier design: QDMTT, IIR, UTPR

Global Minimum Tax (GMT): Implementation

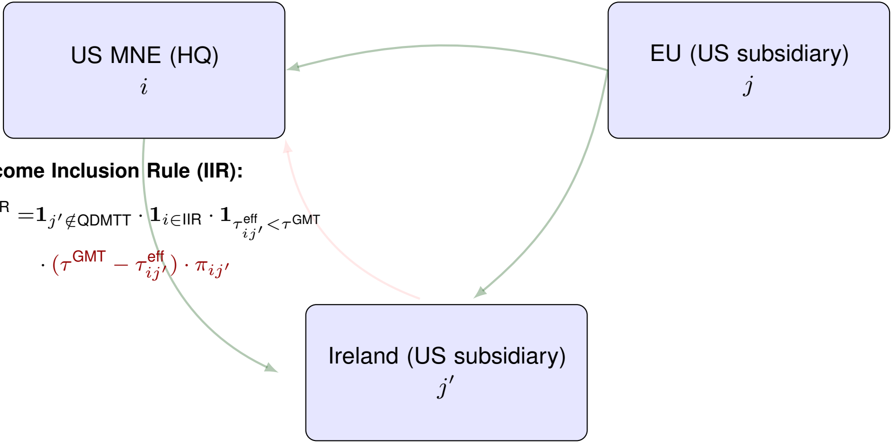


Qualified Domestic Minimum Top-Up Tax (QDMTT):

$$T_{ij'}^{\text{QDMTT}} = \mathbf{1}_{j' \in \text{QDMTT}} \cdot \mathbf{1}_{\tau_{ij'}^{\text{eff}} < \tau^{\text{GMT}}} \cdot (\tau^{\text{GMT}} - \tau_{ij'}^{\text{eff}}) \cdot \pi_{ij'}$$

Carve-out

Global Minimum Tax (GMT): Implementation



Income Inclusion Rule (IIR):

$$T_i^{\text{IIR}} = \mathbf{1}_{j' \notin \text{QDMTT}} \cdot \mathbf{1}_{i \in \text{IIR}} \cdot \mathbf{1}_{\tau_{ij'}^{\text{eff}} < \tau^{\text{GMT}}} \cdot (\tau^{\text{GMT}} - \tau_{ij'}^{\text{eff}}) \cdot \pi_{ij'}$$

Qualified Domestic Minimum Top-Up Tax (QDMTT):

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Carve-out

Global Minimum Tax (GMT): Implementation

US MNE (HQ)
i

EU (US subsidiary)
j

Income Inclusion Rule (IIR):

$$T_i^{\text{IIR}} = \mathbf{1}_{j' \notin \text{QDMTT}} \cdot \mathbf{1}_{i \in \text{IIR}} \cdot \mathbf{1}_{\tau_{ij'}^{\text{eff}} < \tau^{\text{GMT}}} \cdot (\tau^{\text{GMT}} - \tau_{ij'}^{\text{eff}}) \cdot \pi_{ij'}$$

Undertaxed Profits Rule (UTPR):

$$T_{ij}^{\text{UTPR}} = \mathbf{1}_{j' \notin \text{QDMTT}} \cdot \mathbf{1}_{i \notin \text{IIR}} \cdot \mathbf{1}_{\tau_{ij'}^{\text{eff}} < \tau^{\text{GMT}}} \cdot \mathbf{1}_{j \in \text{UTPR}} \cdot s_{ij} \cdot (\tau^{\text{GMT}} - \tau_{ij'}^{\text{eff}}) \cdot \pi_{ij'}$$

Ireland (US subsidiary)
j'

Qualified Domestic Minimum Top-Up Tax (QDMTT):

$$T_{ij'}^{\text{QDMTT}} = \mathbf{1}_{j' \in \text{QDMTT}} \cdot \mathbf{1}_{\tau_{ij'}^{\text{eff}} < \tau^{\text{GMT}}} \cdot (\tau^{\text{GMT}} - \tau_{ij'}^{\text{eff}}) \cdot \pi_{ij'}$$

Carve-out

Corporate Tax Reforms and Profit Shifting return

- Before TCJA, a worldwide tax system:

$$\lambda_{LT} = 1 - \exp\left(-\frac{\left(\tau_{US}^{\text{pre}} - \left(\iota_{LT}\tau_{US}^{\text{pre}} + (1-\iota_{LT})\tau_{LT}\right)\right)(1-\varphi_{i,LT})}{\left(1-\tau_{US}^{\text{pre}}\right) W_{US}\,\psi_{i,LT}}\right)$$

$\rightarrow$ Before TCJA, repatriation of foreign income is subject to US taxation of τ_{US}^{pre}.

$\rightarrow$ Assume a repatriation rate of $\iota_{LT} \in [0,1]$ for profits booked in LT.

Corporate Tax Reforms and Profit Shifting return

- After TCJA, without GILTI and FDII:

$$\lambda_{LT} = 1 - \exp\left(-\frac{(\tau_{US} - \tau_{LT})\,(1 - \varphi_{i,LT})}{(1 - \tau_{US})\,W_{US}\,\psi_{i,LT}}\right)$$

→ The territorial shift removes the repatriation residual $\iota_{LT}\left(\tau_{US}^{\text{pre}} - \tau_{LT}\right)$ from the destination rate.

→ the reduction in τ_{US} decreases λ_{LT} as it shrinks the tax differential.

Corporate Tax Reforms and Profit Shifting return

- With only GILTI:

$$\lambda_{LT} = 1 - \exp\left(- \frac{\left(\tau_{US} - \left(\left(1 - \chi^{GILTI}\right)\tau_{US} + \left(1 - \chi^{FTC}\right)\tau_{LT}\right)\right)\left(1 - \varphi_{i,LT}\right)}{\left(1 - \tau_{US}\right) W_{US}\, \psi_{i,LT}} \right)$$

$\rightarrow \lambda_{LT} \nearrow$ in χ^{GILTI} and χ^{FTC}: GILTI raises the tax rate of income in LT to 10.5%–13.125 %

Corporate Tax Reforms and Profit Shifting return

- With only FDII:

$$\lambda_{LT} = 1 - \exp\left(-\frac{\left(\left(1-\chi^{FDII}FDR\right)\tau_{US}-\tau_{LT}\right)\left(1-\varphi_{i,LT}\right)}{\left(1-(1-\chi^{FDII}FDR)\tau_{US}\right)W_{US}\,\psi_{i,LT}}\right)$$

$\rightarrow$ λ_{LT} $\searrow$ in FDII rate χ^{FDII}: FDII decreases the tax rate of foreign income

Corporate Tax Reforms and Profit Shifting return

- With only GMT:

$$\lambda_{LT} = 1 - \exp\left(- \frac{(\tau_{US} - \tau_{GMT})\,(1 - \varphi_{i,LT})}{(1 - \tau_{US})\,W_{US}\,\psi_{i,LT}} \right)$$

$\rightarrow$ λ_{LT} $\searrow$ in τ_{GMT}: subsidiaries of US MNEs pay top-up tax in LT to 15%

Corporate Tax Reforms and Profit Shifting return

- With TCJA+GMT under a "side-by-side" system:

$$\lambda_{LT} = 1 - \exp\left(-\frac{(\tau_p^* - \tau_{LT}^*)(1 - \varphi_{i,LT})}{(1 - \tau_p^*)W_{US}\,\psi_{i,LT}}\right)$$

$\rightarrow$ $\tau_p^* = (1 - \chi_{\text{FDII}} FDR)\,\tau_{US}$

$\rightarrow$ $\tau_{LT}^* = \max\left\{\left(1 - \chi^{FTC}\right)\tau_{LT} + \left(1 - \chi^{GILTI}\right)\tau_{US},\ \tau_{\text{GMT}}\right\}$: the effective GILTI rate or the GMT floor, whichever binds

The changes to GILTI and FDII by OBBBA return

OBBBA rewrote the rules for both GILTI and FDII significantly, effective for tax years after December 31, 2025.

The goal of the OBBBA was to broaden the tax base by eliminating tangible asset exemptions:

- GILTI is now NCTI (Net CFC Tested Income): increases the ETR from 10.5% to 12.6%
  - → removes the QBAI exclusion of tangible assets
  - → reduce the tax deduction for foreign income from 50% to 40%
  - → increase the foreign tax credits from 80% to 90%
- FDII is now FDDEI (Foreign-Derived Deduction Eligible Income): increases the ETR on eligible export income from 13.125% to 14%
  - → removes the QBAI exclusion of tangible assets
  - → reduces the tax deduction for export income from 37.5% to 33.34%.
  - → excludes interest and R&D expenses from being allocated against DEI, which increases the tax subsidy for R&D intensive firms
- OBBBA also made 100% bonus depreciation permanent (§168(k)); our TCJA experiment's permanent 100% expensing matches post-OBBBA law

Bonus depreciation in the model `return`

- The rental cost of tangible capital is only partially deductible from the tax base:

$$\tilde{\pi}_{ij} \;=\; \pi_{ij} + (1 - b_j)\, r_j k_{ij}, \qquad j \neq TH$$

  where $b_j = e_j \times \theta_j$ is the effective expensing rate: statutory bonus rate $\times$ eligible investment share (Barro and Furman 2018)

- TCJA: e: $0.50 \to 1.00$ and θ: $0.80 \to 0.66$ (eligibility narrows) $\Rightarrow$ b_{US}: $0.40 \to 0.66$ (BEA fixed-asset tables)

- When $b = 1$, the corporate tax falls on pure profits only: no effect on the user cost of tangible capital

- The QBAI is computed on an ADS (straight-line) basis, so it is unaffected by bonus depreciation

- OBBBA (2025) made 100% expensing permanent $\Rightarrow$ our permanent-TCJA experiment matches current law

Comparing the QBAI in TCJA and carve-out in GMT return

Characteristic	QBAI (GILTI)	QBAI (FDII)	GMT Carve-Out
Carve-out rate	10% of tangible assets	10% of tangible assets	5% of tangible assets + 5% of payroll
Exclusion from base	10% $\times$ QBAI subtracted from GILTI income	10% $\times$ QBAI subtracted from FDII-eligible income	5% $\times$ net book value of assets + 5% $\times$ payroll
Tax rate on residual	$0.5 \times 21\% = 10.5\%$	$(1 - 0.375) \times 21\% = 13.125\%$	Top-up to 15% $(15\% -$ ETR)
Blending	Across all CFCs (not per country)	Across all domestic operations	No — per country (no cross-country offset)

- The residual income after the QBAI exclusion faces a higher tax rate than the GMT carve-out in productive, low-tax jurisdictions ($15\% - 11.4\% = 3.6\%$ for LT in our calibration).
- Thus, the GILTI's QBAI mechanism provides greater tax relief than the GMT carve-out.

Valuing the SBIE in units of capital: $0.05K + 0.05\,wL \approx 0.06K$ at $\alpha = 0.25$: a smaller shield than QBAI's $0.10K$, applied at the (lower) top-up rate.

Subject to Tax Rule (STTR) and SBTI Safe Harbor return

- **Subject to Tax Rule (STTR)**:
  - → Treaty-based Pillar 2 instrument, separate from the GloBE rules (QDMTT/IIR/UTPR)
  - → Source countries may levy a 9% withholding tax on intra-group royalties, interest, and fees paid to recipients taxed below 9%
  - → Targets developing countries unable to implement the full GloBE framework; operates via the OECD MLI

- **Substance-Based Tax Incentive Safe Harbor (SBTI-SH)**:
  - → Qualifying substance-based tax incentives are added back to covered taxes when computing the GloBE ETR, shielding them from triggering a top-up
  - → The tax reduction from a qualifying incentive is treated as if it were a tax payment, preserving the ETR above 15% without raising the statutory rate

Measuring profit shifting in the model

Taking the Model to the Data Return

- The profits shifted out of region j by firm ω is

$$ps_{ij}(\omega) = \pi_{ij}^{NS}(\omega) - \pi_{ij}(\omega).$$

  where π_{ij}^{NS} are the profits a firm would have reported in region j if it did not shift profits ($\lambda_{LT} = \lambda_{TH} = 0$).

- $\pi_{ijt}^{NS}(\omega)$ can be computed in PE or in GE
  - $\rightarrow$ we use the PE calculation which correspond to the conceptual framework in Tørsløv et al. (2023)

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- $\pi_{ijt}^{NS}(\omega)$ can be computed in PE or in GE
  - $\rightarrow$ we use the PE calculation which correspond to the conceptual framework in Tørsløv et al. (2023)
- Aggregating firm-level shifted profits yields the total profits shifted out of region j:

$$PS_{jt} = \sum_{i=1}^{I} \int_{\Omega_{ij}} ps_{ijt}(\omega)\, d\omega.$$

Taking the Model to Data

Assigned parameters and target moments

Statistic	US	Europe	Low-tax	RoW	Tax haven
Population (NA = 100)	100	137	17	2,041	–
Real GDP (NA = 100)	100	98	18	383	–
Corporate tax rate τ_j (%; non-U.S.: TWZ)	28.9	17.3	11.4	17.4	3.3
GAAP EATR U.S. non-MNEs (%)	26.0	–	–	–	–
GAAP EATR U.S. MNEs (%)	25.6	–	–	–	–
Foreign MNEs' VA share (%)	11.12	19.82	28.73	9.55	–
Total lost profits (% GDP)	0.8	1.2	–	1.4	–
Lost profits to TH (%)	66.4	44.5	–	71.1	–
Chg. agg. lost profits after TCJA (%)	-48.1	–	–	–	–
Imports from... (% GDP)					
US	–	1.54	0.33	8.92	–
Europe	1.01	–	2.99	8.24	–
Low tax	1.49	12.43	–	7.89	–
Row	2.36	3.70	0.59	–	–

return

Calibrated Parameters

Parameter value	US	Europe	Low-tax	RoW	Tax haven
TFP (A_i)	1.00	0.76	1.21	0.24	–
Elast. of substitution betw. varieties (ϱ)	5	5	5	5	–
Prod. dispersion (η_i)	4.69	4.72	5.19	4.56	–
Utility weight on leisure (ψ_i)	1.47	1.46	1.45	1.45	–
Intangible share (ϕ)	0.10	0.10	0.10	0.10	–
Fixed export cost (κ_i^X)	3.2e-3	6.7e-3	1.8e-3	2.7e-2	–
Variable FDI cost (σ_i)	0.44	0.54	0.51	0.54	–
Fixed FDI cost (κ_i^F)	1.98	2.71	0.82	13.5	–
High-tax repatriation share (ι_H)	0.73	–	–	–	–
Low-tax repatriation share (ι_L)	0.75	–	–	–	–
Cost of shifting profits to LT ($\psi_{i,LT}$)	1.17	0.42	–	3.19	–
Cost of shifting profits to TH ($\psi_{i,TH}$)	0.84	1.34	–	2.24	–
Fixed FDI cost to TH ($\kappa_{i,TH}$)	0.04	0.09	–	0.85	–
Variable export cost (ξ_{ij}) from ...					
US	–	3.04	3.25	1.72	–
Europe	2.15	–	1.72	1.34	–
Low tax	2.27	1.57	–	1.55	–
RoW	2.29	2.58	3.06	–	–

return

Counterfactual Analysis

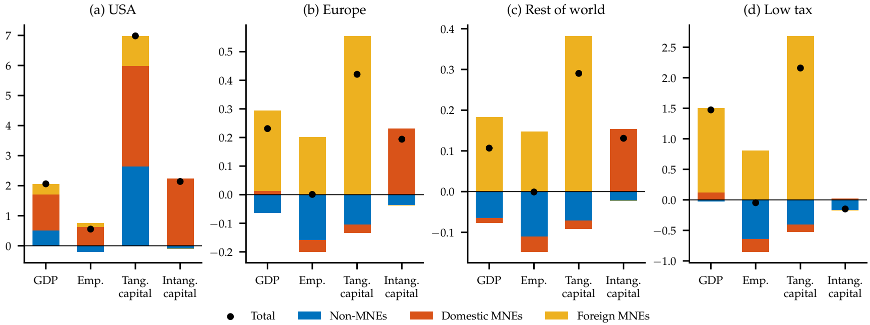
Macro effects of TCJA: full decomposition return

Region	Lost profits/ MNE profits	CIT rev.	GDP	Emp.	Tangible capital	Intangible capital
(a) Territorial system						
USA	29.64	-20.90	0.92	1.08	-0.16	7.46
Europe	-0.02	-0.14	0.21	0.02	0.53	-0.44
Rest of world	-0.01	-0.09	0.06	0.01	0.36	-0.33
Low tax	–	9.31	0.48	-0.20	0.78	-0.93
(b) = (a) + statutory tax cut						
USA	16.74	-32.83	1.50	0.74	3.60	3.47
Europe	0.03	0.10	0.24	0.01	0.45	0.12
Rest of world	0.00	0.15	0.11	0.01	0.31	0.07
Low tax	–	4.40	0.54	-0.10	0.70	0.06
(c) = (b) + more accelerated depreciation						
USA	17.47	-39.57	2.69	0.95	8.08	3.31
Europe	0.04	0.28	0.24	0.01	0.45	0.16
Rest of world	0.01	0.33	0.11	0.00	0.31	0.10
Low tax	–	4.63	0.54	-0.10	0.69	0.13

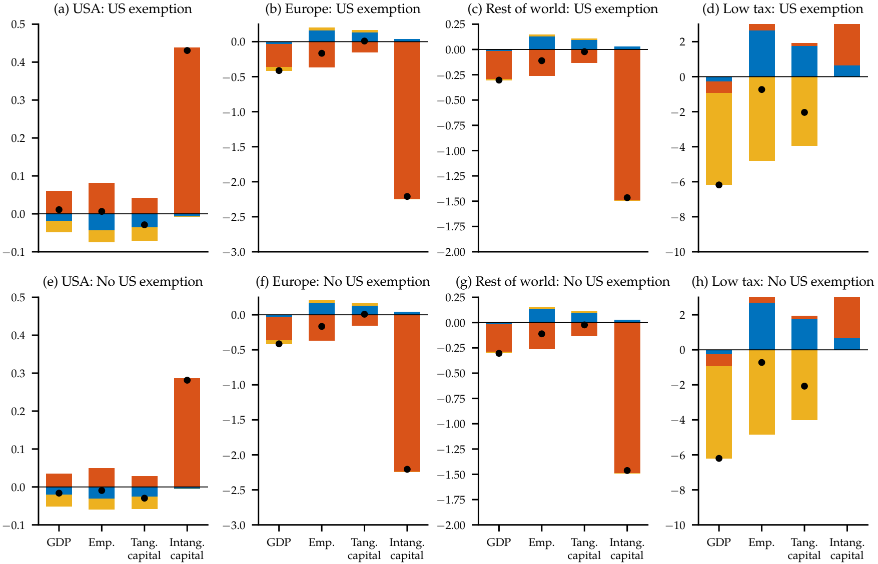
Region	Lost profits/ MNE profits	CIT rev.	GDP	Emp.	Tangible capital	Intangible capital
(d) = (b) + GILTI						
USA	-3.33	-27.35	1.49	0.42	3.66	2.97
Europe	0.02	0.08	0.23	0.01	0.44	0.10
Rest of world	0.00	0.13	0.10	0.00	0.30	0.06
Low tax	–	2.72	1.48	-0.06	2.18	-0.31
(e) = (b) + FDII						
USA	14.37	-32.68	0.97	0.63	2.71	2.69
Europe	0.04	0.05	0.24	0.01	0.44	0.16
Rest of world	0.01	0.10	0.11	0.00	0.30	0.11
Low tax	–	3.42	0.54	-0.08	0.69	0.16
(f) Full TCJA						
USA	-4.02	-34.53	2.06	0.56	6.98	2.14
Europe	0.04	0.20	0.23	0.00	0.42	0.19
Rest of world	0.01	0.25	0.11	-0.00	0.29	0.13
Low tax	–	1.97	1.47	-0.05	2.16	-0.15

Percent changes relative to the pre-TCJA steady state; lost profits in pp. of MNE profits. Panels: (a) territorial only; (b) + rate cut; (c) (b) + bonus depreciation; (d) (b) + GILTI; (e) (b) + FDII; (f) full TCJA.

The effects of TCJA by firm type return



Macro effects of the GMT by firm type return



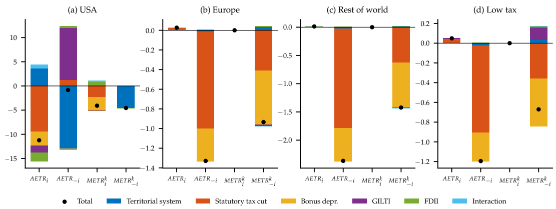
Macro effects of the GMT: full results return

Region	Lost profits/ MNE profits	CIT rev.	GDP	Emp.	Tangible capital	Intangible capital
(a) Complete US exemption						
USA	0.07	-0.48	0.01	0.01	-0.03	0.43
Europe	-9.63	3.02	-0.41	-0.17	0.01	-2.21
Rest of world	-3.76	1.39	-0.30	-0.11	-0.02	-1.47
Low tax	–	6.02	-6.18	-0.73	-2.04	4.99
(b) US MNEs subject to QDMTT						
USA	-0.61	-0.18	-0.02	-0.01	-0.03	0.28
Europe	-9.63	3.02	-0.41	-0.17	0.01	-2.21
Rest of world	-3.76	1.39	-0.30	-0.11	-0.02	-1.46
Low tax	–	5.57	-6.20	-0.72	-2.08	5.02

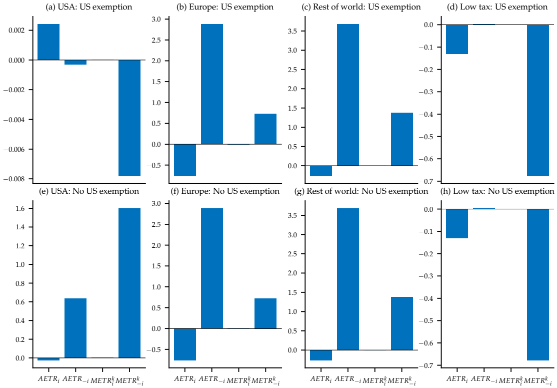
Percent changes relative to the post-TCJA steady state; lost profits in pp. of MNE profits.

Statutory vs. effective tax changes: AETR and METR return

(a) TCJA



(b) GMT



The TCJA moves average and marginal effective rates for U.S. MNEs by several points; the GMT moves them by basis points and only for MNEs in adopting regions.

Alternative calibrations and experiments return

- PV-of-MACRS measure of expensing: $\Delta b \approx 1/5$ of baseline $\Rightarrow$ larger role for the rate cut, smaller for bonus depreciation; combined effect slightly smaller $\Rightarrow$ expensing still generates more output than the headline-rate cut in this experiment
- Tax-equivalent of the non-rate provisions: an 11 pp. statutory hike restores pre-TCJA profit shifting $\Rightarrow$ territorial $+$ GILTI $+$ FDII $+$ expensing are jointly worth $\approx$ 5 pp. of corporate tax
- Lower profit-shifting target (TWZ scaled by -40%): both reforms' effects shrink; the TCJA's barely change
- FDI technology spillovers ($\upsilon > 0$): ripple effects amplified: TCJA's on Europe $\times$4 and RoW $\times$8; the GMT's effect on the U.S. up to $\times$50 spillover technology
- Labor supply: inelastic labor and GHH preferences bracket the log-utility baseline

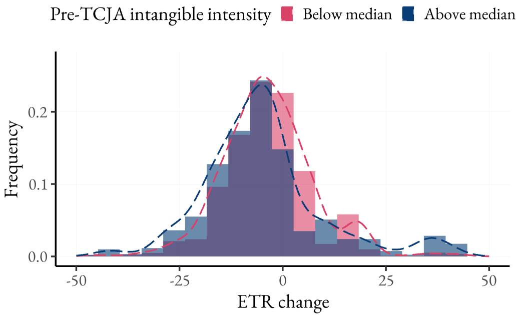
Empirical Study of TCJA

Empirical design return

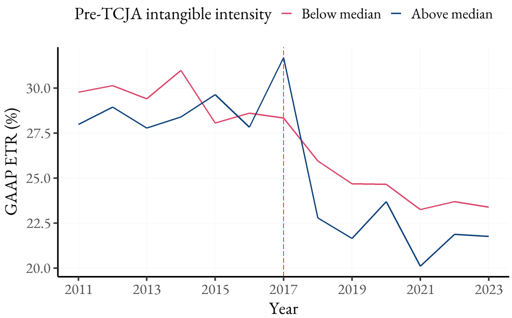
- Sample: Compustat North America 2012–22; U.S. firms with positive pre-tax income in 2016
  - $\rightarrow$ Consolidated statements: sales, tangible/intangible capital, investment, pre-tax income, tax liabilities
  - $\rightarrow$ Intangible capital following Peters and Taylor (2017); MNE status: a foreign subsidiary (WRDS Company Subsidiary data) *and* nonzero foreign income or taxes in the preceding three years
- Treatment: apply the post-TCJA rules to each firm's 2016 characteristics: the 21% rate on domestic income and the 10.5% GILTI rate on foreign pre-tax income; scale the change in liability by 2016 pre-tax income
  - $\rightarrow$ mean simulated cut: 10.2 pp. of pre-tax income, with wide dispersion within industries
- Timing: fiscal 2017–18 only partially treated (blended rates under IRC §15; bonus applies after Sep 27, 2017), so responses are read from fiscal 2019 onward
- Identification: firm FE $+$ industry $\times$ MNE status $\times$ year FE; weighted by 2015–16 revenue; firm-clustered SEs
- Extensive margin: no differential opening or closing of foreign subsidiaries by more-exposed firms $\Rightarrow$ consistent with inframarginal FDI in the model

Firm-Level Effects of the TCJA: The Role of Intangible Intensity

Supplementary descriptive evidence; outside the paper (sample through 2023 vs. the paper's 2022 cutoff).



(a) Histogram of ETR Changes

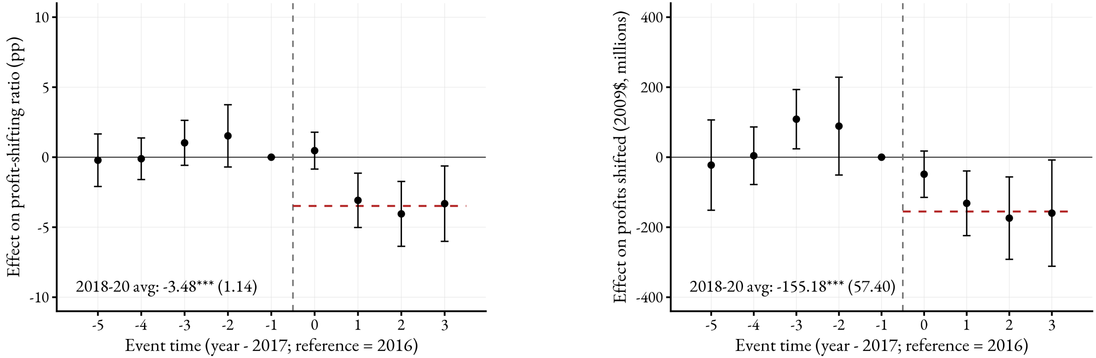


(b) Time Series of ETR

Event study results – Profit shifting `return`

Empirical validation

- We use the panel data on shifted profits constructed by Delis et al. (2024)



(a) Profit-shifting ratio

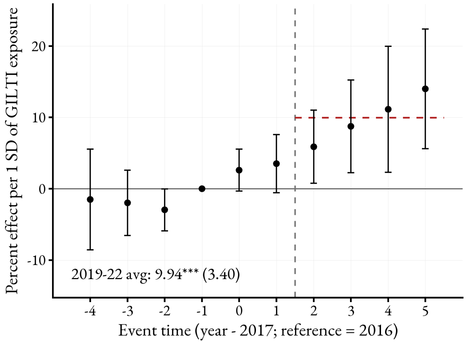
(b) Total profits shifted

2018–20 avg.: shifted-profit share ↓ 3.5 pp., $155m/yr less shifted vs. foreign MNEs in the U.S.

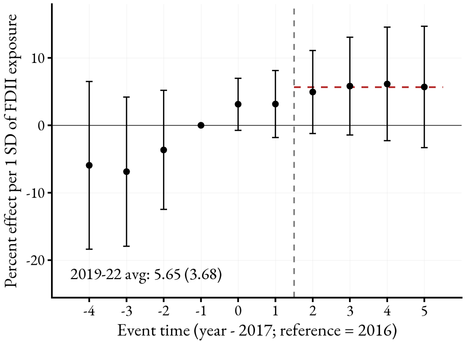
GILTI and FDII exposure results `return`

Event studies: pre-reform (2013–16) exposure to each provision, standardized; tangible capital by jurisdiction from XBRL geographic disclosures.

(a) GILTI exposure → foreign tangible capital



(b) FDII exposure → U.S. tangible capital



- **GILTI**: 1 s.d. more exposure ⇒ foreign PP&E ↑ 9.9% (2019–22 avg.): the QBAI shield
- **FDII**: insignificant among all firms (sales proxy mixes exports & affiliate sales); exporters: ↓ 1.5%/s.d., imprecise ⇒ suggestive

GILTI & FDII: Exposure Construction

GILTI (Global Intangible Low-Taxed Income)

- Taxes foreign income exceeding 10% deemed return on foreign tangible assets (QBAI)
- Incentive: increase foreign PP&E to expand QBAI shield

$$B_i^G = \frac{\max\left\{0,\ \overline{\pi}_i^F - 0.10\,\overline{K}_i^F\right\}}{\overline{A}_i}$$

- $\overline{\pi}^F, \overline{\pi}^D$: foreign/domestic pre-tax income; $\overline{K}^F, \overline{K}^D$: foreign/domestic PP&E; $\overline{A}$: total assets (all 2013–16 averages); exposures standardized to unit variance

FDII (Foreign-Derived Intangible Income)

- 37.5% deduction on foreign-derived income exceeding a 10% return on *domestic* QBAI
- Incentive: higher domestic QBAI *reduces* the benefit $\Rightarrow$ disincentive for domestic tangible investment

$$B_i^F = s_i^F \frac{\max\left\{0,\ \overline{\pi}_i^D - 0.10\,\overline{K}_i^D\right\}}{\overline{A}_i}, \quad s_i^F = \min\left\{1, \frac{\overline{\text{For. Sales}}_i}{\overline{\text{Sales}}_i}\right\}$$